

Designing Subnets

Let total no of addresses granted to the organization be N , & the prefix length is n .

Let the no of addresses assigned to each subnet be N_{sub} & the subnet prefix length is n_{sub} .

Then the following steps need to be carefully followed to guarantee the proper operation of subnetworks.

- ① The no of addresses in each subnetwork should be a power of 2.
- ② The prefix length for each subnetwork should be found using formula $n_{\text{sub}} = 32 - \log_2 N_{\text{sub}}$
- ③ The starting address of each subnet should be divisible by the no of addresses in that subnet. This can be achieved if we assign addresses to larger subnetworks first.

Address Aggregation

One of the advantages of the CIDR strategy is address aggregation (address summarization/route summarization) when blocks of addresses are combined to create a larger block, routing can be done based on the prefix of the larger block.

Special addresses

① Thin-host address (All zero address)

The only address in the block $0.0.0.0/32$ is called the thin-host address (representing itself)

② Limited broadcast address (All ones address)

$255.255.255.255/32$ is used whenever a router or a host needs to send a datagram to all devices in the net.

③ Loopback address

$127.0.0.0/8$: a packet with this destination address will never leave the host, means the ~~host~~ sending host itself is the destination.

④ Private address

$10.0.0.0/8$

$172.16.0.0/12$

$192.168.0.0/16$

$169.254.0.0/16$

The hosts with these addresses can communicate to each other without internet.

This will be discussed in section NAT.

⑤ Multicast address

$225.0.0.0/4$ is reserved for multicast

Dynamic Host Configuration Protocol (DHCP)

DHCP is a application layer program, using the client-server paradigm, that actually helps TCP/IP at the n/w layer.

Address assignment in the organization can be done manually or automatically with: DHCP.

DHCP Message format

0	8	16	24	31
opcode	HType	HLen	HCount	
Transaction ID				
Time Elapsed		Flags		
Client IP Address				
Your IP Address				
Server IP Address				
Gateway IP Address				
Client hardware address				
Server Name				
Boot file Name				
Options				

Fields:

- ① opcode : operation code, request (1) & reply (2)
- ② HType : Hardware type (Ethernet, ...)
- ③ HLen : Length of hardware address
- ④ HCount : Maximum no of hops the pkt can travel
- ⑤ Transaction ID : An integer set by the client and repeated by the server.

Time elapsed: no of seconds since the client started to boot

Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used

Client IP address: Set to 0 if client does not know it

Your IP address: The client IP address set by the server

Server IP address: A broadcast address if client does not know it

Gateway IP address: Address of default router.

Server Name: A 64-byte domain name of the server.

Boot file name: A 128-byte file name holding extra info

Options: A 64-byte field with dual purpose. ~~described~~ or ~~kept~~. It can carry either additional info or some specific vendor info

An option is composed of 3 fields: 1-byte tag field
1-byte length field
variable length value field.

There are several tag fields used by vendors. If the tag field is 53, the value field defines one of the 8 msg types as shown below.

Tag	Length	Value
53	1	↓

- | | |
|-----------------|----------------|
| 1. DHCPDISCOVER | 5. DHCPACK |
| 2. DHCPOFFER | 6. DHCPNACK |
| 3. DHCPREQUEST | 7. DHCPRELEASE |
| 4. DHCPDECLINE | 8. DHCPINFORM |

operation 1) DHCP.

